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DIGS 2003 1 Data Visualization for the Humanities

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GitHub: https://github.com/JiaqiGong1/Chicago-crime-dashboard

Chicago Crime Analysis 2025

Data Source

The data used in this project comes from the Chicago Police Department through the City of Chicago Open Data Portal. The dataset, "Crimes - 2001 to Present," contains reported crime records in Chicago from 2001 to the present. This project uses the 2025 data, which includes 235,535 crime records. Each record contains information such as the crime type, date and time, geographic coordinates, whether an arrest was made, and a short description of the incident.

Motivation for the Visualizations

Map: Crime data naturally includes geographic information. A map is the easiest way to show where crimes are concentrated. It helps viewers quickly identify higher-risk areas and compare crime patterns across neighborhoods.

Arrest Rate Bar Chart: Arrest rates vary a lot between different crime types. This may be reflect how difficult the crimes are to investigate. For example, crimes that happen in public or involve immediate police response usually have higher arrest rates, while crimes that are harder to investigate usually have lower arrest rates.

Monthly Trend Line Chart: Crime patterns change throughout the year, and a line chart clearly shows seasonal trends and monthly changes. It helps identify which months have higher or lower crime activity.

Time Bubble Chart: The timing of crimes is important for understanding urban safety patterns. A bubble chart shows both weekday and hour at the same time, making it easier to identify the time periods when crimes happen most often.

Key Findings

- Crime is highly concentrated in specific areas rather than evenly distributed across the city. Most crimes are clustered in the South Side and West Side of Chicago. The downtown area also shows a noticeable concentration of crime, possibly because of higher population density and heavy daily activity.
- Narcotics crimes have the highest arrest rate at 94.8%, followed by Weapons Violations at around 78%. Deceptive Practice has the lowest arrest rate at only 3.0%, a difference of more than 90 percentage points. This suggests that crimes where offenders can be caught immediately usually have much higher arrest rates than crimes that are harder to investigate afterward.
- Crime reached its highest level in July with 22,639 reported cases, while February had the lowest number with 16,505 cases, a difference of 6,134 incidents. Crime levels stayed relatively high from May to October, suggesting that warmer weather and increased outdoor activity may influence crime

frequency.

- The most common crime time was Saturday at 12:00 AM, with 2,591 reported cases. Friday had the highest total number of crimes during the week with 35,268 incidents, while Tuesday had the lowest with 32,058. The safest time of day was between 4:00 AM and 6:00 AM, after which crime activity gradually increased during the afternoon and evening.
- Using the crime-type filter also shows that different crimes have different geographic patterns. For example, narcotics crimes are concentrated in certain neighborhoods, while theft-related crimes appear more often near commercial and downtown areas.

Outlook

If more resources were available, this project could become a real-time dashboard connected to the Chicago crime data API. Users could view crime incidents from the past 24 hours. Social data such as population, income, and school locations could also be added. This would help explore the relationship between crime and social conditions. Historical data could also be used to predict high-risk places and time periods. This could help police plan patrols more effectively.

Statement on Collaboration

This project was completed independently with the help of Claude as an AI assistant during development.

I mainly used Claude for technical support. It helped me build the basic structure of the Dash application, including the page layout and chart setup. It also helped me debug code and understand error messages. When I had ideas for the dashboard design but was not sure how to implement them, Claude helped me improve some layout and styling details.

Most of the important decisions were made independently. The Chicago crime dataset was chosen because I live in Chicago. I also decided which charts to use, how to organize the dashboard, and how to explain the findings. The dark theme and orange highlight colors were selected to improve readability and match the topic of crime data.

Overall, I am satisfied with the result. The project successfully created an interactive dashboard using more than 235,000 real crime records. During the process, I solved many technical and design problems. Working with Claude helped me turn my original ideas into a complete dashboard project.